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EXPERIMENT 11

Explore Tor network for anonymous communication

AIM:
To understand the working of the Tor network, its role in maintaining anonymity, and how it is used for secure and private communication over the internet.

PROCEDURE:

1. Log in to the TryHackMe platform and start the Tor room.
2. Study the basic concepts of the Tor network and how it provides anonymity using layered encryption.
3. Understand how data is routed through multiple nodes (onion routing) to hide user identity.
4. Explore the structure of the Tor network including entry nodes, relay nodes, and exit nodes.
5. Learn how to access and use the Tor Browser for anonymous browsing.
6. Analyze the advantages and limitations of using Tor for privacy and security.
7. Complete the tasks in the room and record observations about how Tor ensures anonymity.

Tor

A beginner orienteered guide on using the Tor network

45 min 25,850

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Room completed (100%)

Task 1 Unit 1 - Tor

Task 2 Unit 2 - Proxychains

Task 3 Unit 3 - Tor browser

Task 1 Unit 1 - Tor

Tor is a free and open-source software for enabling anonymous communication. Tor directs Internet traffic through a free, worldwide, volunteer overlay network consisting of more than seven thousand relays to conceal a user's location and usage from anyone conducting network surveillance or traffic analysis. Using Tor makes it more difficult to trace Internet activity to the user: this includes "visits to Web sites, online posts, instant messages, and other communication forms". Tor's intended use is to protect the personal privacy of its users, as well as their freedom and ability to conduct confidential communication by keeping their Internet activities unmonitored.

In penetration testing, there might be a need to conduct a full-fledged black-box test. This is a form of testing in which security professionals have to deal with such things as firewalls and other mechanisms of restriction on the customer's side. In this case, the Tor network can be used in order to constantly change IP and DNS addresses and therefore successfully overcome any restrictions.

In this unit, we are going to install the Tor service and learn basic commands.

Answer the questions below

Run `apt-get install tor` to install/update your Tor packages.

No answer needed — ✓ Correct Answer

Run `service tor start` to start the Tor service.

No answer needed — ✓ Correct Answer

Run `service tor status` to check Tor's availability.

No answer needed — ✓ Correct Answer

Run `service tor stop` to stop the Tor service.

No answer needed — ✓ Correct Answer

Task 3 Unit 3 - Tor browser

Tor browser, as seen from its name, is a browser that transfers all its traffic through TOR and by using firefox headers makes all Tor users look the same.

On a daily basis, Tor browser is useful for anyone who wants to keep their internet activities out of the hands of advertisers, ISPs, and web sites. That includes people getting around censorship restrictions in their country, police officers looking to hide their IP address or anyone else who doesn't want their browsing habits linked to them.

Install Tor browser on your system (It is not necessarily to do this on your Kali Machine).

Windows, Mac OS installation

Kali Linux guide

Answer the questions below

Finish the installation.

No answer needed — ✓ Correct Answer

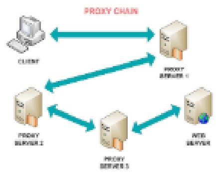
Launch the Tor Browser and set your privacy settings to Level 2 (Safer).

No answer needed — ✓ Correct Answer

What is the search engine at the following onion address: https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/

DuckDuckGo — ✓ Correct Answer

Task 2 Unit 2 - Proxychains



https://github.com/haad/proxychains

Proxychains - a tool that forces any TCP connection made by any given application to follow through proxy like TOR or any other SOCKS4, SOCKS5 or HTTP(S) proxy.

Proxychains is widely used by pentesters during the reconnaissance stage (For example with nmap).

Answer the questions below

Let's start with running `apt install proxychains` to install/update proxychains tool.

No answer needed ✓ Correct Answer

Now it's time to configure proxychains to work properly.

Run `nano /etc/proxychains.conf` to edit the settings. (Note: You can use any text editing tool instead of nano.)

No answer needed ✓ Correct Answer

We can now see, that most of the methods are under comment mark. You can read their description and decide on using one of them in the future. For this lesson let's uncomment `dynamic_chain` and comment others (simply put `#` to the left). Additionally, it is useful to uncomment `proxy_dns` in order to prevent DNS leak. Scroll through the document and see whenever you want to add some additional proxies at the bottom of the page (which is not required at this point).

Apply all the settings.

No answer needed ✓ Correct Answer

Now let's check our settings.

Start the TOR service and run `proxychains firefox`. Usually, you are required to put `proxychains` command before anything in order to force it to transfer data through Tor.

No answer needed ✓ Correct Answer

After the Firefox has loaded, check if your IP address has changed with any website that provides such information. Also, try running a test on `dnsleaktest.com` and see if your DNS address changed too.

NOTE: All other web browser windows should be closed before opening Firefox through proxychains!

No answer needed ✓ Correct Answer

RESULT:

Successfully understood the concept of the Tor network, its working mechanism, and its importance in ensuring anonymity and privacy in online communication.